Journal of Dynamics and Games

doi:10.3934/jdg.2026016

A NOVEL APPROACH OF DOCTORS' HANDWRITTEN MEDICAL PRESCRIPTION RECOGNITION USING A DEEP NEURAL NETWORK AND GENERATING DIGITAL RECORDS OF THE PATIENTS

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Abstract. Although handwritten prescriptions are still frequently used in medical practice, automated recognition is severely hampered by their irregular spacing, variety of handwriting styles, and use of non-standard abbreviations. Current OCR techniques frequently rely on printed text, are insufficiently scalable to handle recently approved drug names, and are restricted to small or single-language datasets. The main research question arises from this gap: How can a multilingual, scalable, and highly accurate framework for the trustworthy recognition of handwritten medical prescriptions be developed? To close this gap, we suggest a hybrid deep learning OCR system that combines multilayer perceptrons (MLPs) and convolutional neural network (CNN) layers. Through GPU-accelerated mini-batch training, the model can be incrementally adjusted, is computationally efficient, and generalizes to various handwriting styles. The framework provides a reliable, scalable, and extremely accurate method for integrating handwritten prescriptions into electronic health records (EHR) systems through digitization. It was trained on a large dataset that contained medical, numerical, and textual elements.

1. **Introduction.** In order to guarantee proper medication administration and to direct treatment, handwritten medical prescriptions continue to be an essential part of clinical practice. Medication mistakes brought on by unreadable handwriting, however, are frequent and can be harmful, especially for patients with severe conditions. For patient safety and efficient healthcare delivery, prescription interpretation must be accurate and automated. Current computer vision and optical character recognition (OCR) techniques face difficulties due to the inherent variability and complexity of handwritten medical notes [1, 2, 3, 4]. Non-standard abbreviations, irregular spacing, and inconsistent handwriting styles significantly reduce the reliability of traditional OCR systems in clinical settings. To address these problems, our analysis proposes an OCR framework driven by deep learning.

2020 *Mathematics Subject Classification.* Primary: 68T10; Secondary: 68U10, 62P10.

Key words and phrases. Optical character recognition (OCR), convolutional neural networks (CNN), deep neural networks (DNN), multilayer perceptrons (MLP), electronic health records (EHR), and handwritten prescription recognition.

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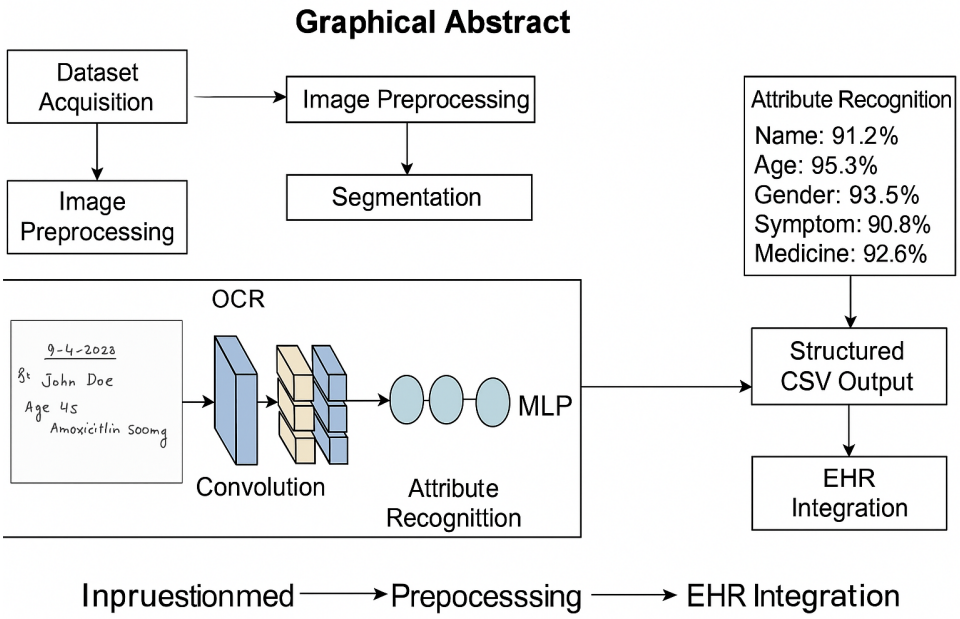


FIGURE 1. Graphical Abstract

Names, ages, symptoms, and information about prescription drugs are just a few of the patient-specific data that this framework can consistently extract and transform into structured digital records. The framework combines segmentation-guided text analysis, an LSTM-CTC module for precise sequence recognition, and Convolutional Neural Networks (CNNs) with max-pooling layers and ReLU activation functions for effective feature extraction. The CSV format in which the extracted data is exported facilitates the management of reliable electronic health records (EHRs) and enables additional applications such as illness prediction and clinical decision support. Because the extracted data is saved in a CSV format, it is simpler to administer a reliable electronic health record (EHR) and perform follow-up tasks like clinical decision support and illness prediction. In addition to digitization, the method [5] uses intelligent fuzzy decision-support mechanisms based on Sugeno-Weber triangular norms and advanced interval-valued intuitionistic fuzzy (CIVIFS) systems. This combination enhances reliability, interpretability, and adaptability in healthcare management, including critical applications like COVID-19 treatment centers. The proposed framework integrates AI-driven fuzzy decision support with OCR-based prescription recognition to improve the accuracy of medical records and enable scalable, data-driven clinical decision-making. **Motivation:** Through the urgent need to improve EHR accuracy, decrease medication errors, and integrate handwritten medical data with state-of-the-art artificial intelligence and fuzzy decision-support systems, this research advances intelligent, data-driven healthcare management.

2. **Literature review:** The increasing need for Big Data and Machine Learning solutions has caused them to be widely used in many different fields [6]. Digital technologies are revolutionising healthcare management and security, according to

recent research [7, 8, 9]. Over the past few decades, continuous research in healthcare digitisation has established the sector as a key pillar of the global economy. To improve clinical decision-making, diagnostic precision, and the overall standard of patient care, researchers have demonstrated a growing interest in integrating Artificial Intelligence (AI) into healthcare systems.

2.1. Keyword spotting in Doctors' handwritten medical prescriptions: Roy et al. presented a keyword-spotting approach for information retrieval for handwritten medical prescriptions in a June 2017 study published in the International Journal of Expert Systems with Applications. Their method optimised classification accuracy for handwritten medical records by combining a Tandem Feature framework based on Multilayer Perceptrons (MLP) with a Hidden Markov Model (HMM). Additionally, this study presented a new retrieval technique for recognising the names of medications and illnesses in doctors' handwritten prescriptions [10].

Limitations:

- Primarily relies on prescription sections that are printed rather than handwritten.
- Performance can be affected by unusual fonts, noise, or irregular formatting.
- Lacks robustness when handling fully handwritten prescriptions.
- has limited scalability for large datasets.
- Does not support the recognition of multilingual prescriptions.

2.2 Medicine Box: Doctor's prescription recognition using deep learning:

This system leverages deep learning to interpret physicians' prescriptions on medication boxes [11] and, similar to a mobile app, relies on a Convolutional Neural Network (CNN) to accurately recognize handwritten drug names.

Limitations:

A portion of the sample data set is used, and the remainder is not.

2.3. Recognition of handwritten medical prescription using signature verification :

A study by S. Rani *et al.* [12] in *Computational and Mathematical Methods in Medicine* (September 2022, e-Collection 2022) used machine learning models, such as Support Vector Machines (SVM) and Naive Bayes Classifier, to identify handwritten medical prescriptions using signature verification techniques.

Limitations:

The model is very large and has a lot of time complexity.

2.4. Transformer-based handwritten recognition for medical prescription applications:

In a 2019 paper presented at the University of Waterloo and the University of Strathclyde's joint Virtual Colloquium, Z. Raisi *et al.* [13] suggested a transformer-based architecture for handwritten text recognition that uses two-dimensional image representations to maintain spatial continuity in data.

Limitations:

Inconsistencies in the dataset used in this study had an impact on the overall performance and reliability of the model.

2.5. Devnagari numerical recognition through combining several connectionist Classifiers' decisions: A paper by R. Bajaj *et al.* was published in

Sadhana – Academy Proceedings in Engineering Sciences, 2002. In order to recognize Devanagari numerals, [14] suggested a multi-classifier connectionist architecture that combines several feature-based classifiers. Using a range of characteristic features as classification inputs, the model's accuracy was 89.6%.

Limitations:

- Due to their capacity to recognize sequential and spatial patterns, Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) typically perform exceptionally well on image-based recognition tasks.
- Conversely, multilayer connectionist models are more likely to overfit, particularly when they are trained on small datasets.

2.6. Integrating various feature extraction methods for handwritten Devnagari character recognition:

An OCR framework for handwritten Devanagari characters was proposed by S. Arora *et al.* [15] in a paper presented at the *Third International Conference on Industrial and Information Systems* and the *IEEE Region 10 Conference (TENCON)* in 2008. The system used a neural classifier to identify basic character symbols and four Multilayer Perceptron (MLP)-based classifiers for decision-level fusion to improve classification accuracy. Both the structural and statistical properties of the handwritten symbols were used for feature extraction to increase robustness. These features were then combined using a multi-classifier ensemble approach.

Limitations:

- The method was limited to Devnagari characters and could not generalise to other alphabetic scripts.
- It was unable to process multilingual handwritten datasets, limiting its use across different languages.

2.1. **Contributions and insights obtained from the literature review.** A careful review of previous studies on the recognition of handwritten prescriptions reveals steady improvements in algorithmic design and model performance. However, there are still a lot of unanswered questions in areas like data scalability, cross-linguistic adaptability, and computational efficiency. The following discussion, which uses the *what–how–why* framework to outline methodological innovations and their associated limitations, summarizes key studies.

(i) **Roy *et al.* (2017): What:** proposed a technique for extracting relevant entities from handwritten prescriptions by identifying keywords. **How:** To improve drug and disease name recognition, a Hidden Markov Model and a Multilayer Perceptron-based tandem feature extraction module were used. **Why:** Highlighted the benefit of integrating sequential and spatial modelling; however, the framework was limited to semi-printed text and did not generalize to handwritten datasets in multiple languages.

(ii) **Kamalanaban (2019): What:** Developed a lightweight convolutional neural network for classifying handwritten content on medicine labels. **How:** Incorporated convolutional feature maps in a mobile-optimized architecture to achieve faster inference suitable for real-time processing. **Why:** Demonstrated the practicality of on-device prescription recognition but showed reduced robustness under varied handwriting styles and illumination conditions due to limited training data.

(iii) **S. Rani *et al.* (2022): What:** Explored conventional machine-learning algorithms, including SVM and Naïve Bayes, for handwritten prescription

interpretation combined with signature verification. **How:** Fused textual and biometric features to enhance authentication reliability. **Why:** Higher identification accuracy on small datasets, but deployment in large clinical repositories was limited by high computational overhead.

(iv) **Z. Raisi *et al.* (2019): What:** Presented a transformer-based architecture for medical handwriting decoding. **How:** Used two-dimensional positional embeddings to preserve spatial dependencies while decoding the sequence. **Why:** Outperformed recurrent-based models in contextual learning;

(v) **R. Bajaj *et al.* (2002): What:** Created a multi-classifier connectionist method for Devanagari numeral recognition in handwriting. **How:** Several feature-driven classifiers were integrated to capture temporal and spatial relationships in handwritten numbers. **Why:** Obtained an accuracy of 89.6%, offering preliminary proof of ensemble learning; however, its ability to generalize was constrained by the small dataset size.

(vi) **S. Arora *et al.* (2008): What:** Suggested a multilayer perceptron classifier-based OCR framework for handwritten Devanagari characters. **How:** used a decision-level ensemble approach to combine structural and statistical descriptors. **Why:** Limited its direct clinical use due to its poor adaptability for multilingual handwritten data, despite achieving improved script recognition.

In essence, the reviewed studies collectively illustrate meaningful advancement in the use of hybrid deep-learning and traditional approaches for handwriting interpretation. Nevertheless, existing models often remain limited by insufficient dataset diversity, low generalization to real-world prescriptions, and computational inefficiency. Addressing these challenges, the present research introduces a unified CNN-based recognition framework optimized for large-scale, multilingual handwritten data—aimed at enhancing recognition accuracy, adaptability, and clinical reliability.

3. Proposed methodology.

3.1. Image acquisition.

In a handwritten prescription recognition system, image acquisition is the first and most important step in the entire Optical Character Recognition (OCR) pipeline. Its main goal is to take a high-quality digital picture of the handwritten document while keeping all the structural and textual information needed for future processing accuracy.

$$I_{\text{input}} = f(x, y, c), \quad \text{where} \quad c \in \{R, G, B\} \tag{1}$$

Prescription photos were gathered for this study from a variety of sources, including pharmacies, healthcare facilities, and hospitals. The suggested recognition model's ability to accurately identify target patterns was demonstrated through the use of a large dataset.

3.1.1. *Different ways in which we captured the images of prescriptions.*

- **Flatbed scanners** We use the flatbed scanner to provide uniform illumination and high-resolution images.
- **Smartphone cameras** For convenience and portability, we used smartphone cameras to acquire images of the prescriptions.

- **Document cameras or webcams** We collected images of prescriptions from hospital web-cams, also for the real-time acquisition in automated hospital record systems.

3.1.2. *Considerations.*

- **Resolution** We considered that a sufficient pixel density is required to preserve fine handwriting details.
- **Lighting Conditions** Even illumination prevents shadows and glare that may obscure strokes.
- **Orientation** Proper alignment ensures that text is horizontally labelled. A minimum skew correction is needed.
- **File formats** Commonly used image formats include JPEG, PNG, and TIFF. Among these, lossless formats such as PNG and TIFF are preferred because they preserve image details without compression loss.

3.1.3. *Edge detection in image preprocessing.* Before the image goes through OCR, edge detection is done to ensure that the entire prescription frame is correctly identified. The method highlights regions of the image that exhibit sudden changes in intensity, usually lining up with the edges of structures or character outlines. One popular method is to calculate the image's gradient magnitude:

$$G = \sqrt{G_x^2 + G_y^2} \tag{2}$$

- G denotes the *overall gradient magnitude* at each pixel, representing the strength of edges.
- G_x is the gradient along the horizontal (x) direction, which records intensity changes from left to right.
- G_y is the gradient along the vertical (y) direction, which records intensity changes from top to bottom. Regardless of direction, the combination of G_x and G_y yields a measure of edge strength. These gradients are usually calculated efficiently using methods such as the Sobel, Canny, or Laplacian operators, which improve the readability and precision of OCR for handwritten prescriptions. The explanation of G_x, G_y, and G can be found in Fig. 2.

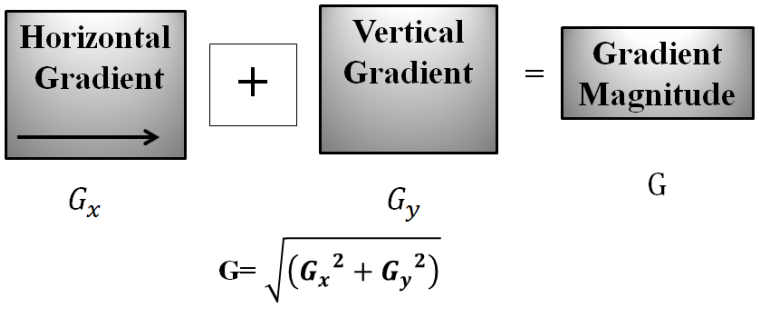


FIGURE 2. Gradient magnitude computation using horizontal and vertical gradients

3.2. **Image pre-processing.**

3.2.1. *Grayscale conversion.* Converts RGB images to grayscale using the standard formula:

$$I_{\text{gray}} = 0.299R + 0.587G + 0.114B \tag{3}$$

This reduces computational complexity while preserving the handwritten structure for further processing in the OCR pipeline. Fig. 3 shows the image after grayscaling.

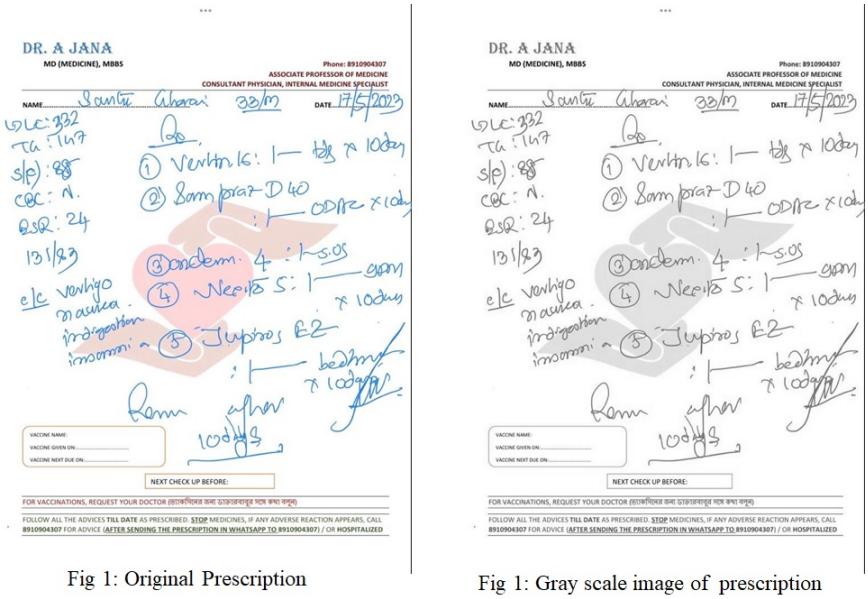


Fig 1: Original Prescription

Fig 1: Gray scale image of prescription

FIGURE 3. Original prescription and its grayscale representation

3.2.2. *Noise removal.* Removes unwanted artefacts such as scanning noise, salt-and-paper noise, or smudges using some filtering processes like Gaussian filtering, Median filtering, and denoising techniques. In our experiments, all applicable filtering techniques were employed to remove noise. Fig. 4 shows the image after noise removal.

3.2.3. *Binarization.* By giving each pixel a value of 0 (black) or 1 (white), binarization transforms a colour or grayscale image into a binary image. In optical character recognition (OCR) systems, this procedure streamlines image data and makes feature extraction easier. The sample image following binarization is displayed in Fig. 5.

$$I_{\text{bin}}(x,y) = \begin{cases} 1, & \text{if } I_{\text{gray}}(x,y) > T \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

Otsu's method, an adaptive and data-driven technique that automatically chooses the ideal threshold by minimizing the intra-class intensity variance between the foreground and background pixels [16], was used in this experiment to determine the threshold value T used for binarization.

$$T = \arg\min_{\tau} \left[\omega_1(\tau)\sigma_1^2(\tau) + \omega_2(\tau)\sigma_2^2(\tau)\right] \tag{5}$$

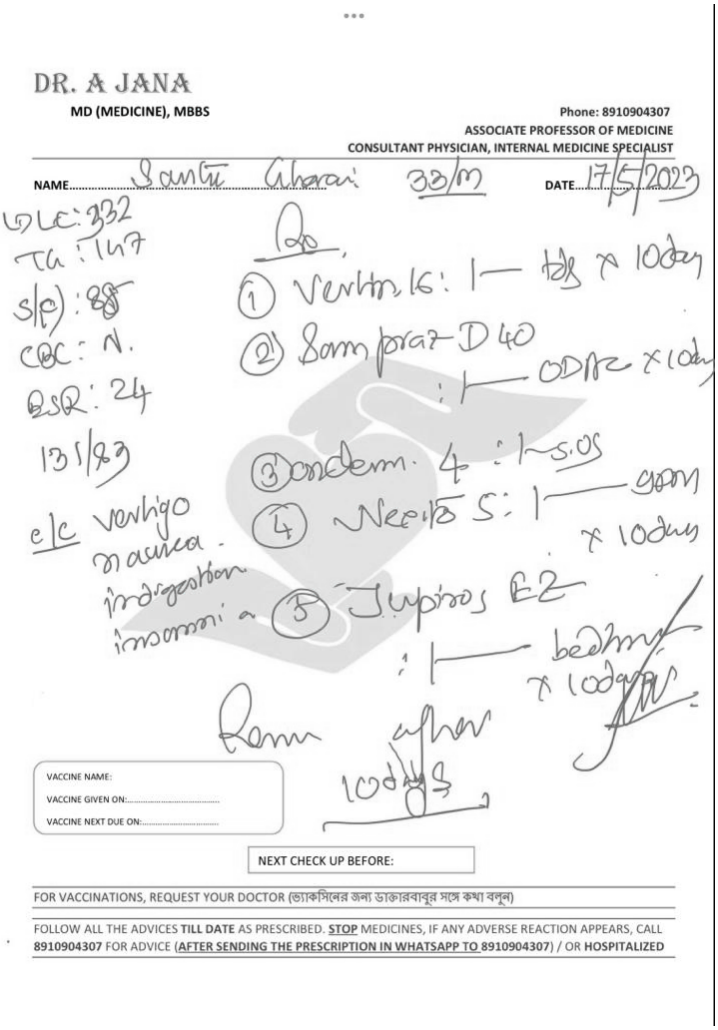


FIGURE 4. Image after noise removal

where $\omega_1(\tau)$ and $\omega_2(\tau)$ denote the probabilities of the foreground and background classes separated by threshold τ, and $\sigma_1^2(\tau)$ and $\sigma_2^2(\tau)$ represent their corresponding variances. Fig. 6 shows the histogram of the binarized prescription image.

3.2.4. *Skew correction.* Skew correction is also called de-skewing. It is a process of detecting and correcting the tilt angle of a scanned or photographed image so that the text or content lines are horizontally aligned. It ensures accurate segmentation, character recognition, and feature extraction in OCR pipelines.

3.2.5. *Morphological operation.* It improves the structure of the text and removes small artefacts. We used Erosion and Dilation in our experiment.

3.2.6. *Contrast enhancement.* Contrast enhancement is a key image pre-processing technique used to improve the visibility of features in an image by increasing the

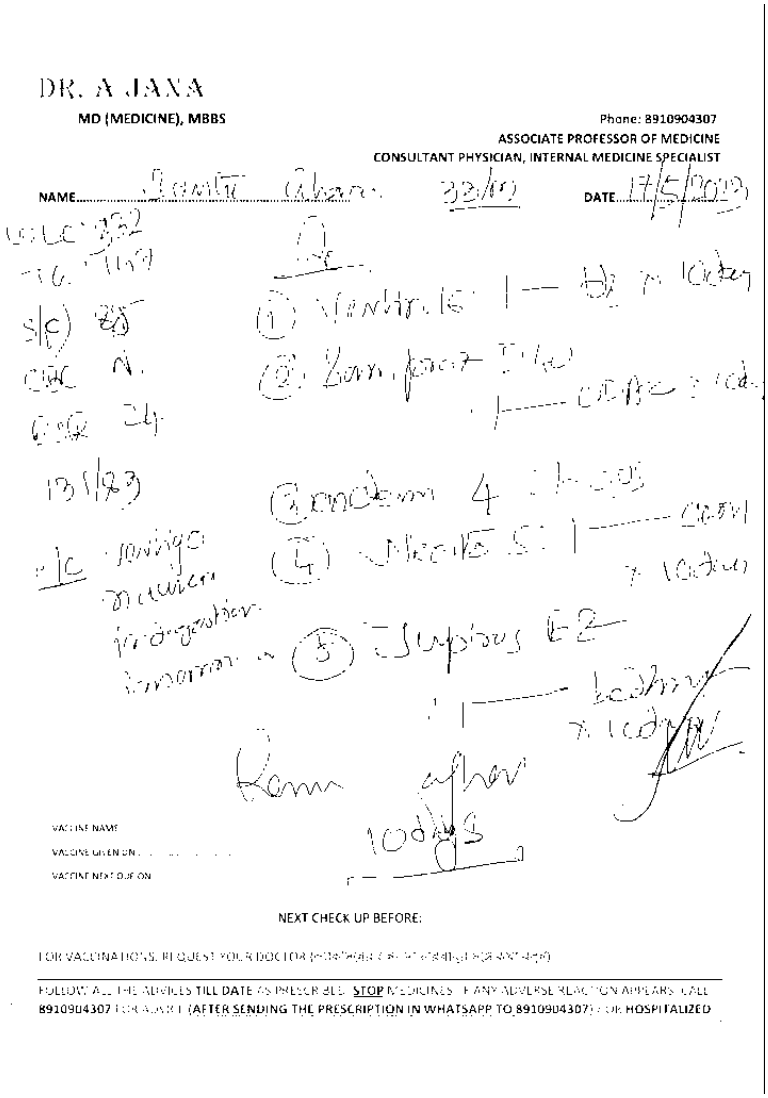


FIGURE 5. Image after binarization

difference between the light and dark regions. In handwritten prescription recognition in our experiment, the contrast enhancement ensures that text written with faint ink or uneven illumination becomes clearer for subsequent Optical Character Recognition (OCR) steps. Fig. 7 shows the sample image after contrast enhancement.

3.3. **Image segmentation.** Text segmentation aims to isolate lines, words, and characters from a handwritten prescription image so that each unit can be processed by an Optical Character Recognition (OCR) model. Accurate segmentation ensures reliable recognition of patient details, symptoms, and medications.

3.3.1. *Phases of image segmentation of collected prescriptions.*

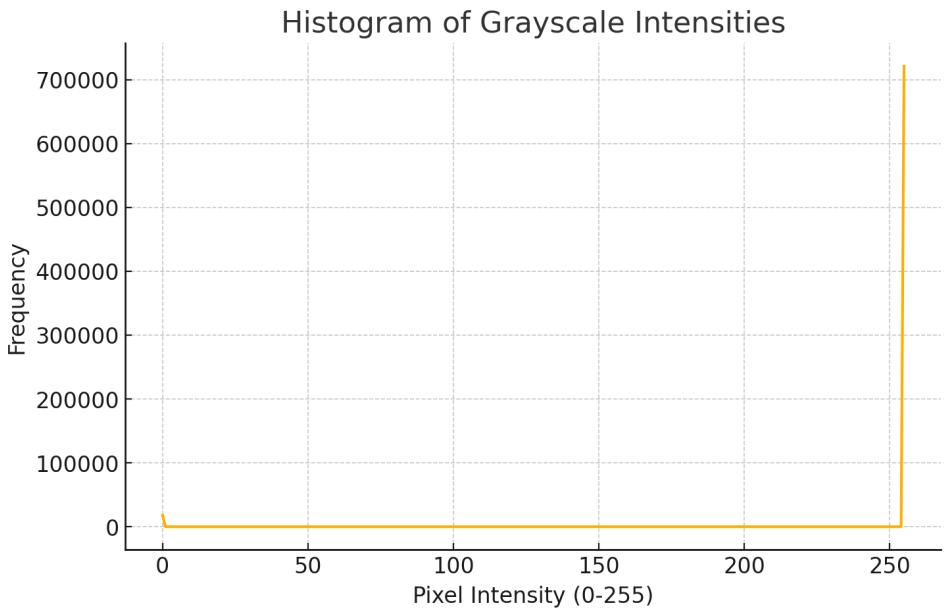


FIGURE 6. Histogram of binarized prescription image

- **Normalisation** In handwritten OCR, a preprocessing method is used to modify scaling, contrast, proportions, and consistency, increasing the dynamic range of the image. Additionally, it modifies brightness levels to preserve consistency and enhance recognition precision. A sample image following normalization is displayed in Fig. 8.
- **Gray scaling** Grayscale transformation is the process of converting an image into different shades of gray, removing color information while maintaining intensity details. Several color space conversion formats can be used to implement this technique, depending on the needs of the particular application.
- **Region of Interest (ROI)** Regions of Interest (ROIs) represent specific portions of an image that are targeted for filtering or processing to extract handwritten text from prescription documents. Initially, all Regions of Interest (ROIs) within the document are identified and isolated for subsequent analysis and recognition.
- **Contour Detection** In image processing, contour detection is employed to extract the boundaries of objects within an image. In handwritten prescription analysis, contour detection facilitates the extraction of the alphabetic character shapes from prescription images. Algorithms such as Sobel, Canny, and Laplacian of Gaussian (LoG) are widely recognised for their effectiveness in contour detection. The contour detection of a sample image is shown in Fig. 9.
- **Cropping and Alphabet Recognition** Cropping is an image processing technique used to remove or adjust the outer boundaries of an image. In Optical Character Recognition (OCR), cropping is applied to isolate specific regions of an image that contain textual information.
- **Printed and handwritten text classification**

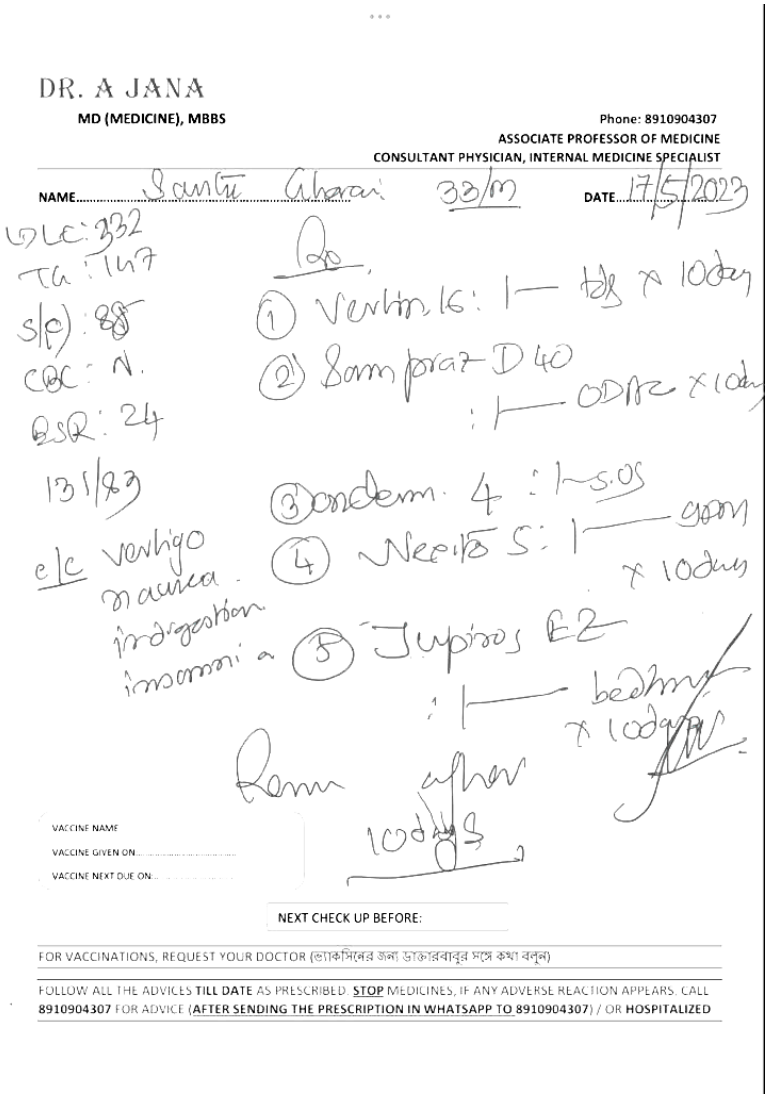


FIGURE 7. Image contrast enhancement

- **Block partitioning**: Tile the image into overlapping or non-overlapping blocks. Overlap of 25–50% helps with boundaries.
- **Feature extraction**: Use a descriptor that separates printed (regular strokes, uniform spacing) from handwriting (irregular strokes, variable texture).
- **Classifier**: Using a classifier, the printed and handwritten texts are classified.

The classification is discussed below in detail.

Figs. 11a–11d present combined images of different sections of a prescription, distinguishing between handwritten and printed components in a sample doctor's prescription.

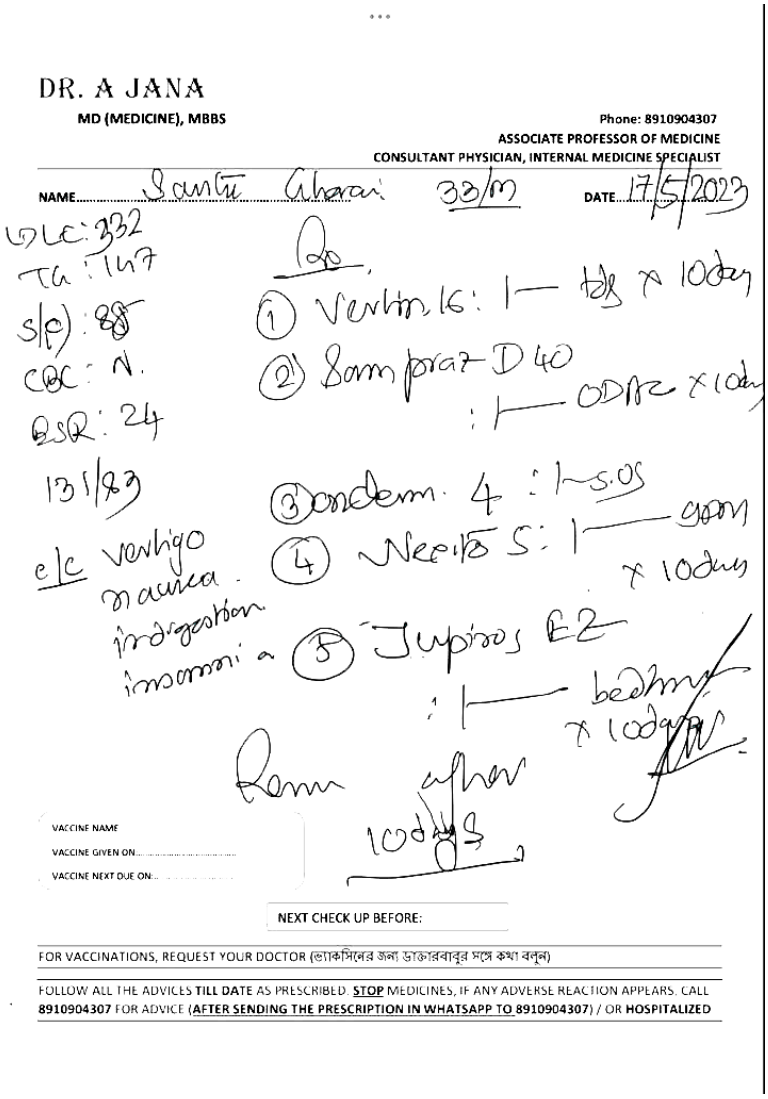


FIGURE 8. Image after normalisation

3.4. **Classifier selection for printed and handwritten text segmentation.** Considering the heterogeneous nature of prescription documents, which contain both machine-printed and manually handwritten regions, a robust image-based classifier was required. After comparative evaluation of multiple models, a hybrid CNN–SVM framework was selected for final implementation. The Convolutional Neural Network (CNN) automatically extracts spatial and texture features that differentiate between the smooth, consistent patterns of printed text and the irregular, stroke-based variations of handwritten content. These learned feature embeddings are then fed into a Support Vector Machine (SVM), which provides sharp decision boundaries for binary classification. This combination achieves high accuracy and stability across a variety of handwriting styles and print fonts by utilizing the representational power of the CNN and the discriminative strength of the SVM.

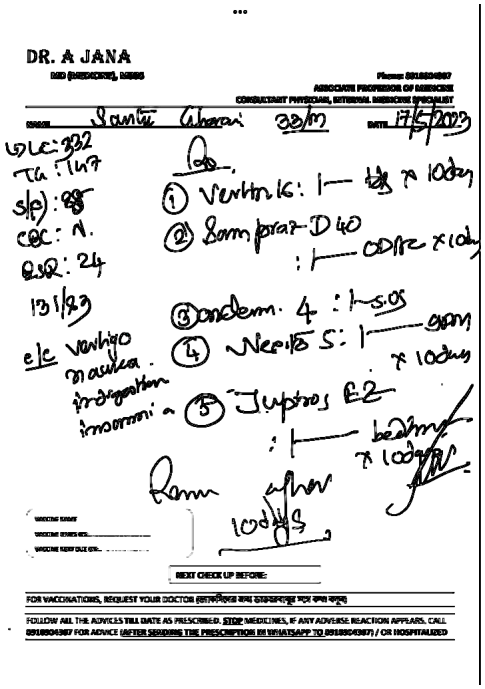


FIGURE 9. Contour detection of an image

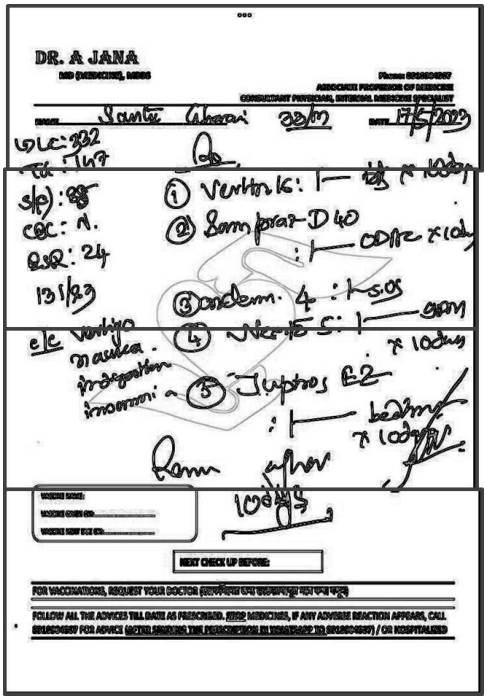
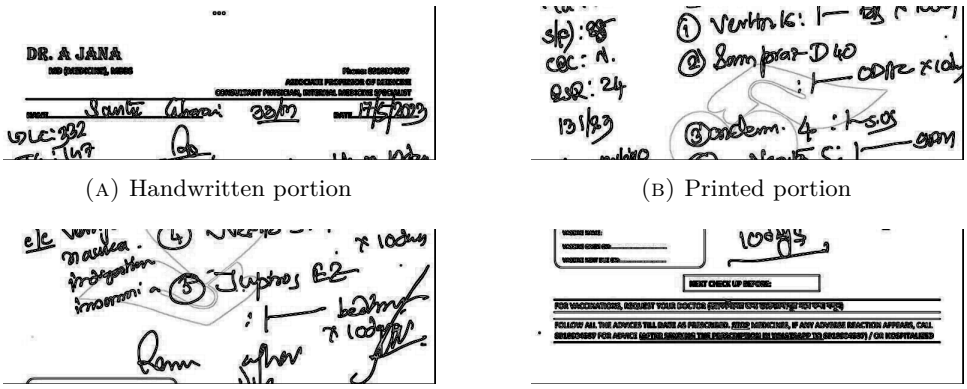


FIGURE 10. Full prescription with contour detection and partitioning



(A) Handwritten portion (B) Printed portion

(C) Mixed content (D) Other handwritten portion

FIGURE 11. Combined images of various sections of a prescription, distinguishing between handwritten and printed components.

3.4.1. *Convolutional neural network (CNN)-based classification.* A Convolutional Neural Network (CNN) classifier was used to improve the recognition accuracy of handwritten and printed text. Because CNNs do not require manual feature extraction, they can automatically learn spatial and textural features directly from raw data, making them very effective for image-based classification tasks.

Advantages

- CNNs autonomously learn discriminative features such as stroke patterns, texture irregularities, and edge orientations.
- They provide robustness to translation, rotation, and scaling variations in handwriting.
- The hierarchical convolutional architecture captures both low-level and high-level visual representations.
- CNNs achieve high accuracy and generalization performance through end-to-end learning.

Classification Steps

1. **Input Image:** The prescription picture with the handwritten and printed text is normalized and resized.
2. **Preprocessing:** Segmentation, contrast normalization, and noise reduction are used.
3. **Convolutional Layers:** Local texture and edge information are extracted by filters.
4. **ReLU Activation:** To learn intricate patterns, non-linearity is introduced.
5. **Pooling Layers:** Essential spatial features are preserved while dimensionality is decreased.
6. **The Fully Connected Layer:** To make decisions, extracted features are processed and flattened.
7. **Output Layer:** Class probabilities for the *Printed* or *Handwritten* categories are generated by softmax activation.

This end-to-end CNN architecture offers high precision and dependability across a variety of prescription formats by efficiently differentiating between printed and handwritten text regions.

3.4.2. *CNN-based handwritten and printed text classification.* The prescription document's printed and handwritten sections were separated using a Convolutional Neural Network (CNN). CNN can perform reliable classification without the need for explicit feature engineering by directly learning texture-based and spatial patterns from image data. Grayscale normalization, noise filtering, and text region segmentation were used as preprocessing techniques on the input prescription image. The CNN model, which consists of several convolutional, pooling, and dense layers, was fed each segmented text patch after it had been resized. The convolutional layers extracted key indicators of handwritten versus printed text, including curvature, stroke width, and edge consistency. The Softmax output layer produced two probability values that matched the classes Printed and Handwritten.

Post-classification, each detected region was visually labeled using colored bounding boxes — blue for printed and green for handwritten text — as shown in Fig. 12. This automated labeling effectively separated structured printed information such as Name, Address, and Date from handwritten components including the patient's details, prescribed medicine, and physician's signature. Fig. 12 illustrates the classification of printed and handwritten text from a handwritten prescription. The blue blocks represent printed text, while the green blocks represent handwritten text after classification. Using the proposed OCR model, the handwritten text is recognised and converted into a digital format. Fig. 12 shows the printed and handwritten classified texts from a handwritten prescription. Blue blocks show the printed, and green blocks show the handwritten texts after classification. In the figure shown below, the blocks show printed and handwritten text separately.

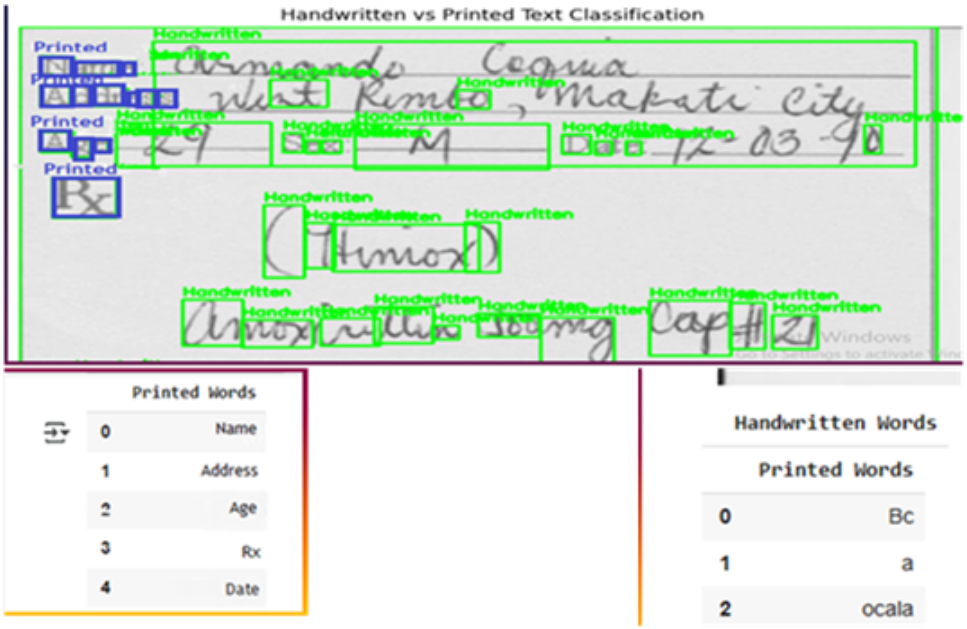


FIGURE 12. Classification of printed and handwritten text regions from a handwritten prescription

Here, some sample texts are also labelled. The printed text is recognised using Python Pytesseract package, and the handwritten text is recognised using our proposed OCR model

TABLE 1. Confusion matrix of CNN-based classifier for printed and handwritten text recognition.

Actual / Predicted	**Printed**	**Handwritten**
Printed	485	15
Handwritten	12	488

Fig. 13 shows the CNN Confusion matrix for printed and handwritten classified texts from a handwritten prescription.

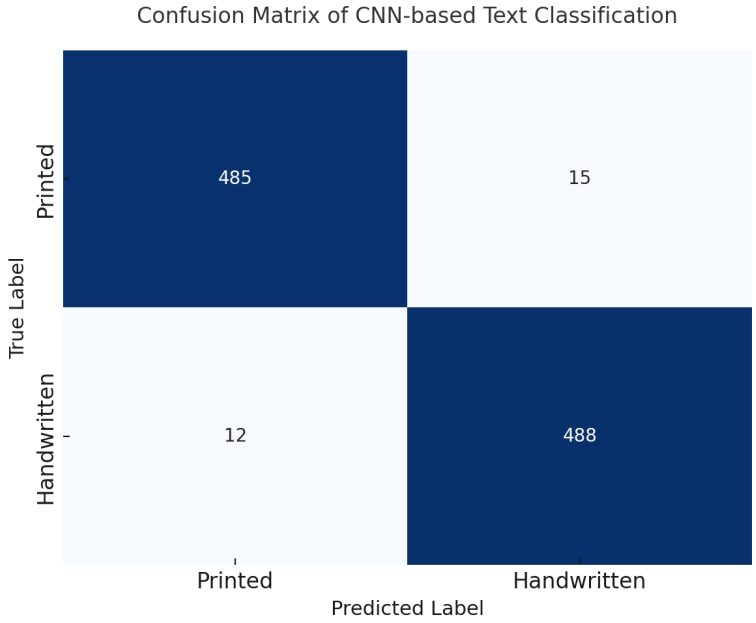


FIGURE 13. Confusion matrix

3.4.3. *Training and evaluation of CNN classifier.* A Convolutional Neural Network (CNN) model was trained to distinguish between printed and handwritten text regions extracted from prescription documents. The dataset comprised a total of 2,000 segmented text patches, obtained after contour-based segmentation of multiple prescription images. Each patch was manually labeled as either *Printed* or *Handwritten* to serve as ground truth for supervised learning. Out of the total dataset, 70% (1,400 samples) was used for training, 15% (300 samples) for validation, and the remaining 15% (300 samples) for testing. The CNN was trained for 50 epochs with a batch size of 32 using the Adam optimizer and categorical cross-entropy loss function. Data augmentation techniques such as random rotation, horizontal shifting, and contrast adjustment were applied to enhance model generalization and prevent overfitting.

3.5. **Read .csv files:** The train, test, and validation datasets are subsequently loaded for processing. In this research, the Python programming language was utilised within the Jupyter Notebook environment. The details of the training dataset, which serves as the primary source of data for model training, are presented in the program. The training process is further carried out using a Convolutional Neural Network (CNN) model to improve the accuracy of handwritten text recognition.

3.6. **Training and testing model:** After the features of the input image are extracted, a preferred Machine Learning algorithm is chosen to extract the perfect features. Different machine learning algorithms, including Convolutional Neural Networks (CNN) and traditional machine learning techniques, are employed to design an OCR system. The choice of an appropriate algorithm depends on the specific application and the dataset's features. Once the model is trained, its performance is evaluated using a separate dataset of handwritten text samples. In the Fig. 14, the flowchart of our proposed methodology is shown.

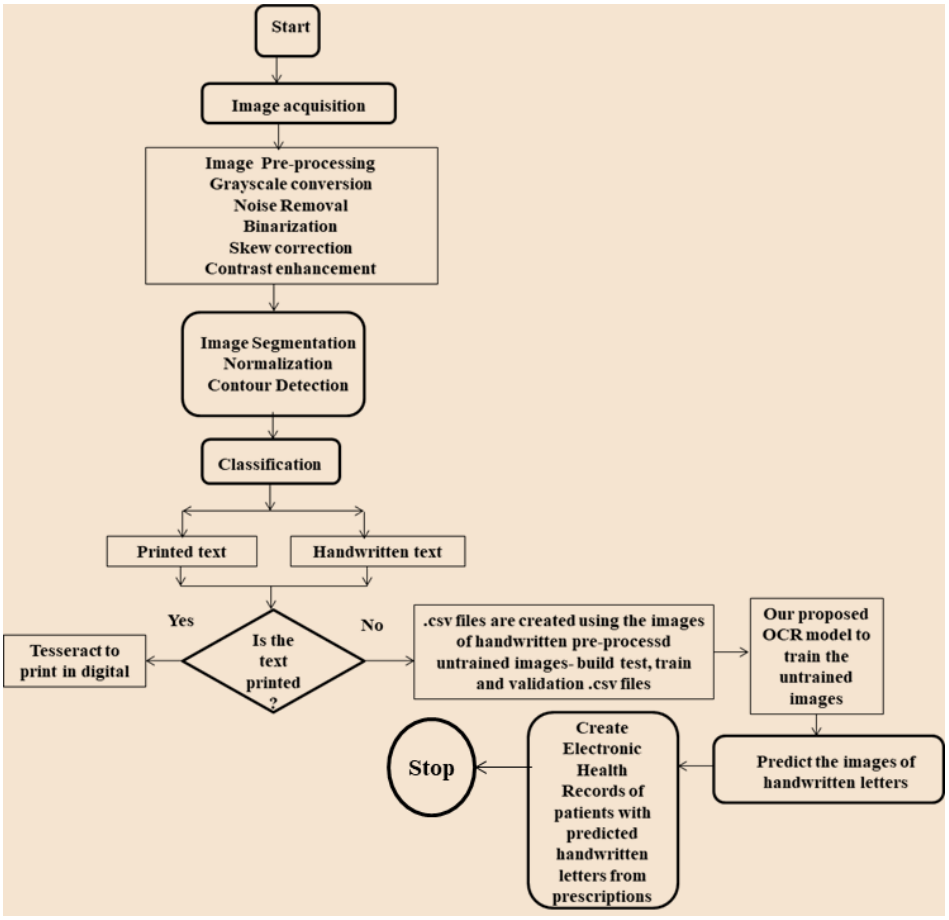


FIGURE 14. flowchart of the methodology

3.7. **System configuration and tools used.** Every experiment was carried out on a high-performance computer running Windows 10 (64-bit) with an Intel Core i7-12700H processor (2.3 GHz, 14 cores), 32 GB DDR5 RAM, and an NVIDIA GeForce RTX 3060 GPU with 6 GB VRAM. OpenCV, NumPy, Pandas, and Matplotlib were utilized for image processing and visualization, while Python 3.11, TensorFlow (v2.x), and Keras frameworks were utilized for model development. Jupyter Notebook was used to conduct the experiments, guaranteeing transparent model tracking, reproducibility, and debugging effectiveness.

3.8. **Dataset collection and preprocessing.** More than 400,000 handwritten prescription samples were gathered from different clinics to create the dataset. Every prescription was scanned at 300 DPI in order to maintain high fidelity and structural details. Normalization, grayscale conversion, and subplots representing the essential characteristics were applied to the scanned images: *Name*, *Age*, *Gender*, *Symptoms*, and *Prescribed Medicines*. Text segmentation and contour detection were used to extract the corresponding regions of interest (ROIs). Fig. 15 displays sample cropped attributes from sample prescriptions.

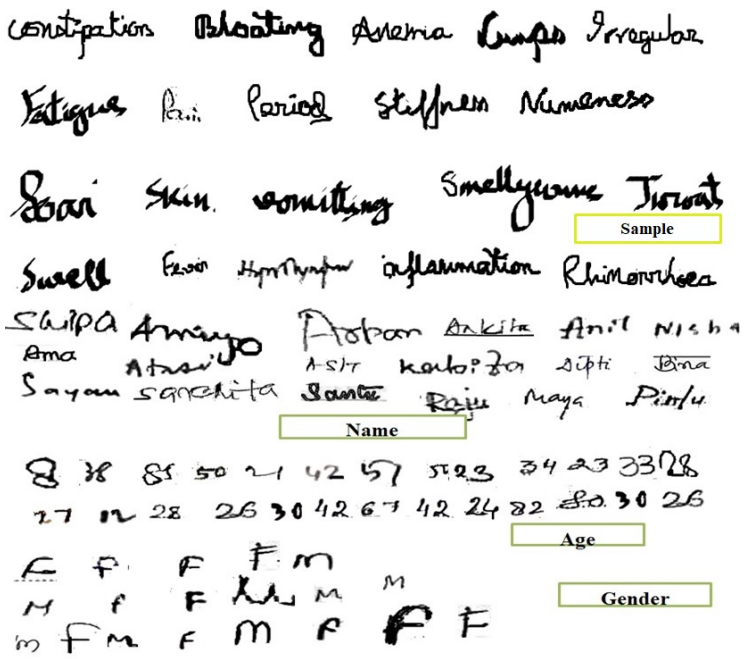


FIGURE 15. Name, Age, and Gender Attributes Extracted from Prescription Samples

3.9. **Creation of CSV files.** The annotated data for each attribute was stored in distinct `.csv` files that included image filenames and matching identity labels. The samples *train.csv*, *test.csv*, and *validation.csv* are all kept exactly as they were used in the experimental setup.

Name test	
File Name	Identity
Tests 1.png	ABDUL
Tests 2.png	ANIL
Tests 3.png	ANKITA
Tests 4.png	ARPAN
Tests 5.png	ASIT
Tests 6.png	ATASI
Tests 7.png	DEBITA
Tests 8.png	DIPTI
Tests 9.png	GOBINDA
Tests 10.png	MOUSUMI
Tests 11.png	MAYA
Tests 12.png	PALLAB
.....	....

Name train	
File Name	Identity
Train 1.png	ABDUL
Train 2.png	ANIL
Train 3.png	ANKITA
Train 4.png	ANNA
Train 5.png	ARPAN
Train 6.png	ASIT
Train 7.png	BINA
Train 8.png	BHATTA
Train 9.png	DEBITA
Train 10.png	PALLAB
Train 11.png	ATASI
Train 12.png	MOUSUMI
.....	....

Name valid	
File Name	Identity
Valid 1.png	ABDUL
Valid 2.png	ANIL
Valid 3.png	ANKITA
Valid 3.png	ARPAN
Valid 4.png	ASIT
Valid 5.png	ATASI
Valid 6.png	DEBITA
Valid 7.png	DIPTI
Valid 8.png	GOBINDA
.....	....

3.10. **Model architecture and training.** The CNN model had about 8,871,354 parameters and was trained over 100 epochs. A Multi-Layer Perceptron (MLP) classifier for label prediction came next. The MLP converted the spatial features—such as edges and strokes—that the CNN had extracted into symbolic representations. Tokenization and image preprocessing are demonstrated in Fig. 16 and Fig. 17.

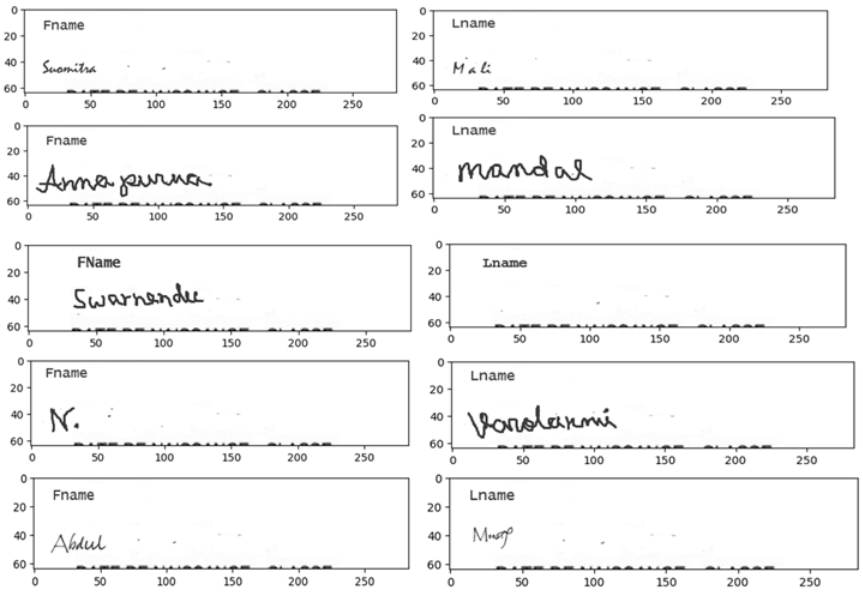
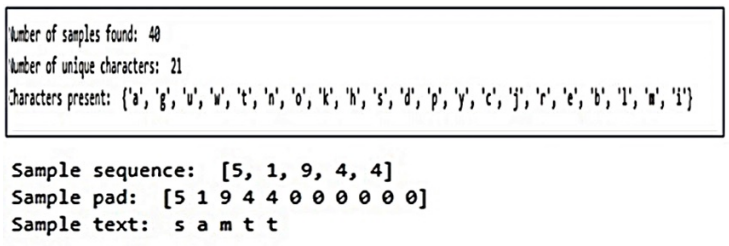
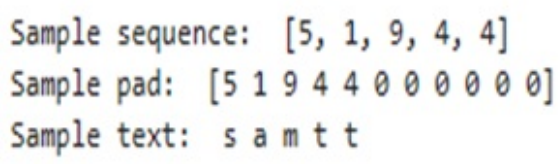


FIGURE 16. Unreadable Image Samples Before Training



(A) Samples Obtained During Training



(B) Sample Sequences After Tokenization

FIGURE 17. Preprocessing and Tokenization of Untrained Images

3.11. **ReLU activation function.** The Rectified Linear Unit (ReLU) activation function was employed for all convolutional and dense layers. It introduces non-linearity by setting all negative input values to zero while passing positive ones unchanged.

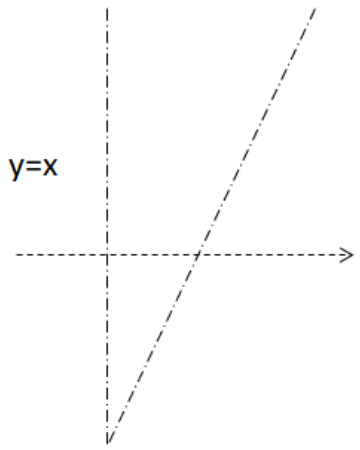


FIGURE 18. Graphical diagram of the ReLU activation function

At $x = 0$, the derivative of the Rectified Linear Unit (ReLU) activation function is undefined due to the presence of a sharp corner, where the function abruptly changes direction from flat (slope = 0) to rising (slope = 1). A derivative represents the instantaneous slope of a function; however, at non-differentiable points such as corners, this slope becomes undefined. For computational simplicity, the ReLU derivative is represented in TensorFlow as either 0 or 1 (predominantly 0), a simplification that has negligible impact on the overall training process. Figure 18 illustrates the graphical representation of the ReLU activation function. The ReLU derivative ensures that positive activations contribute to weight updates, while inactive neurons ($x < 0$) remain frozen.

Let X denote the input image and W_k represent a convolution filter in the layer k. The convolution operation in layer k is defined as

$$Z_k = W_k * X_{k-1} + b_k, \tag{6}$$

where b_k is the bias term. After applying the ReLU activation function, the output activation is given by

$$A_k = \text{ReLU}(Z_k) = \max(0, Z_k). \tag{7}$$

The ReLU behaviour can be summarised as:

- If $Z_k < 0$, the neuron remains inactive.
- If $Z_k > 0$, the neuron propagates the value forward.

During training, the gradient of the loss function L with respect to the weights W_k is computed as

$$\frac{\partial L}{\partial W_k} = \frac{\partial L}{\partial A_k} \cdot \frac{\partial A_k}{\partial Z_k} \cdot \frac{\partial Z_k}{\partial W_k}, \tag{8}$$

where the derivative of the ReLU activation is defined as

$$\frac{\partial A_k}{\partial Z_k} = \begin{cases} 1, & \text{if } Z_k > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

This property enables efficient and stable backpropagation, making ReLU particularly suitable for large-scale OCR networks. The derivative of the ReLU activation function with respect to its input is given by

$$\frac{\partial A_k}{\partial Z_k} = \begin{cases} 1, & \text{if } Z_k > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

A simplified OCR-based CNN pipeline implemented using TensorFlow/Keras is described below:

1. **Input Layer** → Image (grayscale or RGB)
2. **Convolution + ReLU Layers** → Extract feature maps
3. **MaxPooling Layers** → Reduce spatial dimensions
4. **Flatten / Dense Layers with ReLU** → Non-linear feature combination
5. **Output Layer (Softmax)** → Character probabilities

Mathematically, the convolution and activation operations are expressed as

$$Z = W * X + b, \tag{11}$$

$$A = \text{ReLU}(Z), \tag{12}$$

where X denotes the input image or feature map from the previous layer, W is the convolution filter, b is the bias term, Z is the pre-activation output, and A is the activated feature map.

The OCR–CNN processing steps are summarised in Table 3.

TABLE 2. Key computational steps in a CNN-based OCR pipeline with left-aligned equations.

Step	**Equation**	**Role in OCR**
Convolution	$Z_k = W_k * X + b_k$	Detects low-level patterns such as edges and strokes.
ReLU Activation	$A_k = \text{ReLU}(Z_k)$	Retains strong positive features and introduces non-linearity.
MaxPooling	$P_k = \text{MaxPool}(A_k)$	Downsamples feature maps, preserves dominant features, and provides translation invariance.
Next Layer Input	$P_k \rightarrow$ next convolution	Builds higher-level features for character recognition.

Each convolution-ReLU block converts raw pixel intensities into feature maps representing key visual attributes, including edges, corners, and strokes. The Max-pooling layers subsequently downsample these representations, retaining the most discriminative features. Through the hierarchical stacking of such layers, the network develops increasingly abstract representations that facilitate precise recognition of characters and words.

4. **Dataset description.**

4.1. **Name prediction dataset.** The custom dataset for name recognition contained approximately 250,000 handwritten name samples and 15,000 test samples. CNN layers extracted character features, and MLP layers predicted the corresponding class labels. Example results are shown in Fig. 19.



FIGURE 19. Predicted names from handwritten prescriptions

4.2. **Age prediction dataset and algorithms.** To predict handwritten age values, two algorithms were implemented—one for single-digit prediction and another for multi-digit age recognition.

4.2.1. *Algorithm for single-digit prediction (Algo_SID).*

1. Start.
2. Acquire handwritten digit images from prescriptions through scanning, segmentation, and normalization.

3. Generate the required `.csv` files for training, testing, and validation datasets.
4. Import necessary Python libraries and split the datasets into training and testing subsets.
5. Construct a neural network model with four layers using ReLU activation via the Keras Functional API.
6. Configure the input layer with 784 features and hidden layers comprising 300, 100, 100, and 100 neurons, respectively.
7. Train the model for 20 epochs, resulting in 294,210 trainable parameters.
8. Predict single-digit handwritten age values.
9. End.

The single-digit prediction model demonstrated high accuracy in recognizing isolated numerals. However, it exhibited limitations when processing concatenated multi-digit numbers. To address this issue, a more advanced multi-digit prediction approach (Algo_MULTI) was developed, inspired by the methodologies proposed by H.-C. Nuerk *et al.* [17] and Rahman and Almarzooqi [18].

4.2.2. *Algorithm for multi-digit prediction (Algo_MULTI).*

1. Utilize prior single-digit predictions obtained from Algo_SID as input features.
2. Apply the CART algorithm for classification using Decision Tree and Random Forest models [20, 19].
3. Optimize the weights of the MLP with ReLU activation using Stochastic Gradient Descent (SGD).
4. Combine individual digit predictions to form complete multi-digit numbers.
5. Accurately recognize handwritten age values consisting of multiple digits.
6. End.

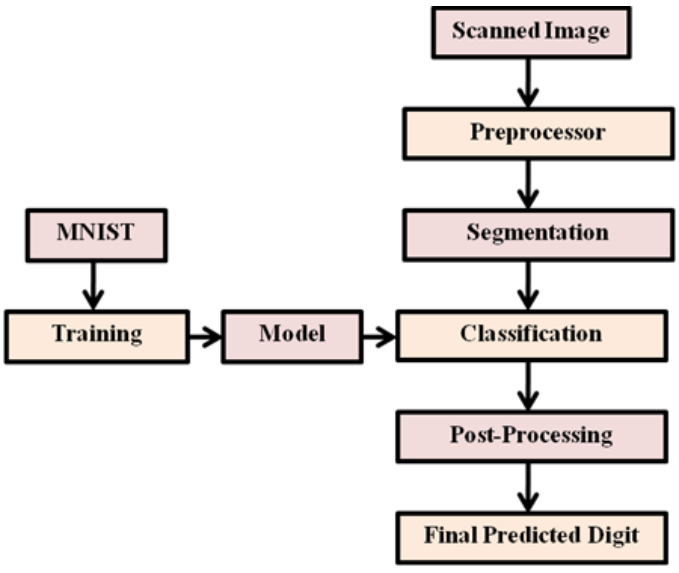


FIGURE 20. Flowchart for multi-digit age prediction

4.2.3. *Finite state machine for digit and white-space segmentation.* Distinguishing adjacent digits from white space is challenging when handwritten numerals are

irregularly spaced. Inspired by Nguyen and Nakagawa [22] and Nadeem *et al.* [21], a Finite State Machine (FSM) was employed to address this issue.

The FSM utilizes run-length thresholds and pixel density information to define three distinct states: *Digit Region*, *White Space*, and *Transition Zone*.

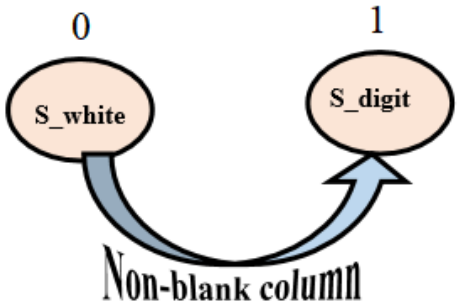


FIGURE 21. State diagram for age prediction using a finite state machine

State transitions are governed by the following rules:

$$\text{Digit} \xrightarrow[\text{low pixel density detected}]{\text{small gap}} \text{White Space}$$

$$\text{White Space} \xrightarrow[\text{stroke appears}]{\text{density increases}} \text{Digit}$$

$$\text{Digit} \xrightarrow[\text{partial stroke observed}]{\text{boundary unclear}} \text{Transition Zone}$$

By effectively separating consecutive digits while ignoring intra-character gaps, This FSM-based segmentation improves the detection of digit boundaries in noisy or irregular handwritten age values.

4.3. **Gender prediction.** A Multilayer Feedforward (MLFF) Neural Network model is used in this study to recognize handwritten alphabets in order to predict gender. One of the most basic types of feedforward artificial neural networks is the MLFF Neural Network, which is made up of several layers of connected perceptrons. Information can only move forward from the initial input to the final output thanks to the network architecture's division into an input section, multiple intermediate processing layers, and an output module. Let $\mathbf{x} = [x_1, x_2, \ldots, x_n]$ be the input vector, and let jth be the hidden layer neuron's output.

$$h_j = f\left(\sum_{i=1}^{n} w_{ij} x_i + b_j\right), \tag{13}$$

where the nonlinear activation function is indicated by $f(\cdot)$ and the synaptic weight and bias associated with the neuron are represented by w_{ij} and b_j, respectively. The final output of the network is computed as

$$y_k = f_o\left(\sum_{j=1}^{m} v_{jk} h_j + c_k\right), \tag{14}$$

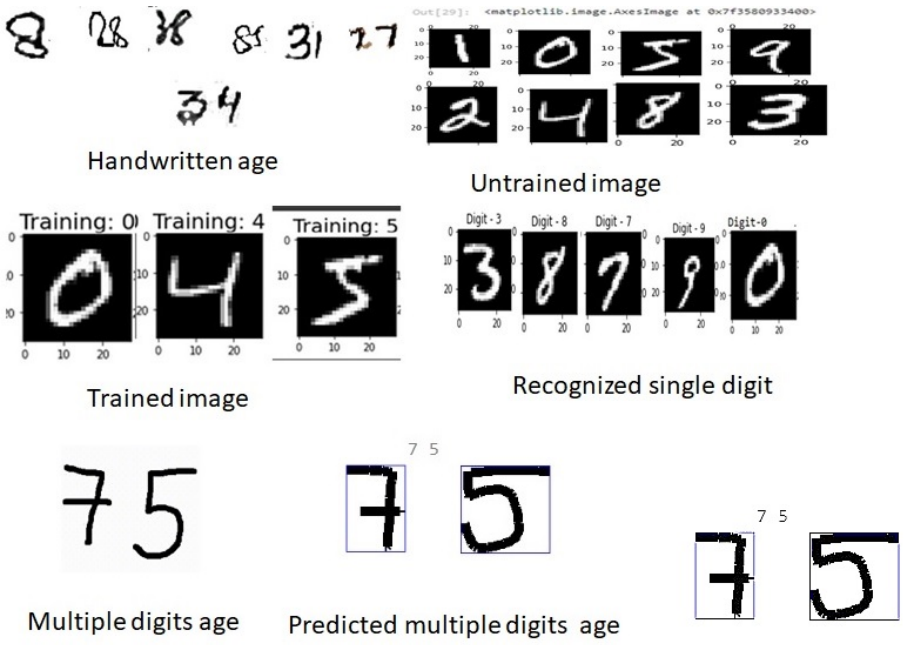


FIGURE 22. Age prediction using FSM-based segmentation

where v_{jk} and c_k denote the output layer weights and biases, respectively, and $f_o(\cdot)$ represents the output activation function. Through the hidden layers, data flows unidirectionally from the inner layers to the outer layers. After each training epoch, this process is iteratively refined to minimise the loss function through back-propagation. The Multilayer Feedforward Neural Network architecture used in this study is illustrated in Fig. 23. The study makes use of a custom dataset assembled from actual patient records in addition to the EMNIST dataset, which comprises 131,600 character images belonging to 47 classes. For training and evaluation, the handwritten gender image data are stored in `.csv` files. The dataset is trained using a Multilayer Perceptron (MLP) as the primary classifier, while a Convolutional Neural Network (CNN) is incorporated to improve label prediction accuracy. The Multilayer Feedforward Network architecture employed in this study is illustrated in Fig. 23.

Each image sample is represented as

$$I(x, y) \in \mathbb{R}^{a \times b}, \tag{15}$$

where a and b denote the image height and width, respectively. The corresponding class labels are defined as

$$y \in \{1, 2, \ldots, C\}, \tag{16}$$

where C represents the total number of gender classes. with $C = 47$ indicating the total number of output categories. For each class C, M_j feature instances are extracted and represented as

$$x_j = \left(x_{j,1},\, x_{j,2},\, \ldots,\, x_{j,M_j}\right),$$

where $j = 1, 2, \ldots, d$, and d denotes the total number of features. For each sub-feature vector in a given class, the mean $\mu_{j,n}$ and variance $\sigma^2_{j,n}$ are calculated to

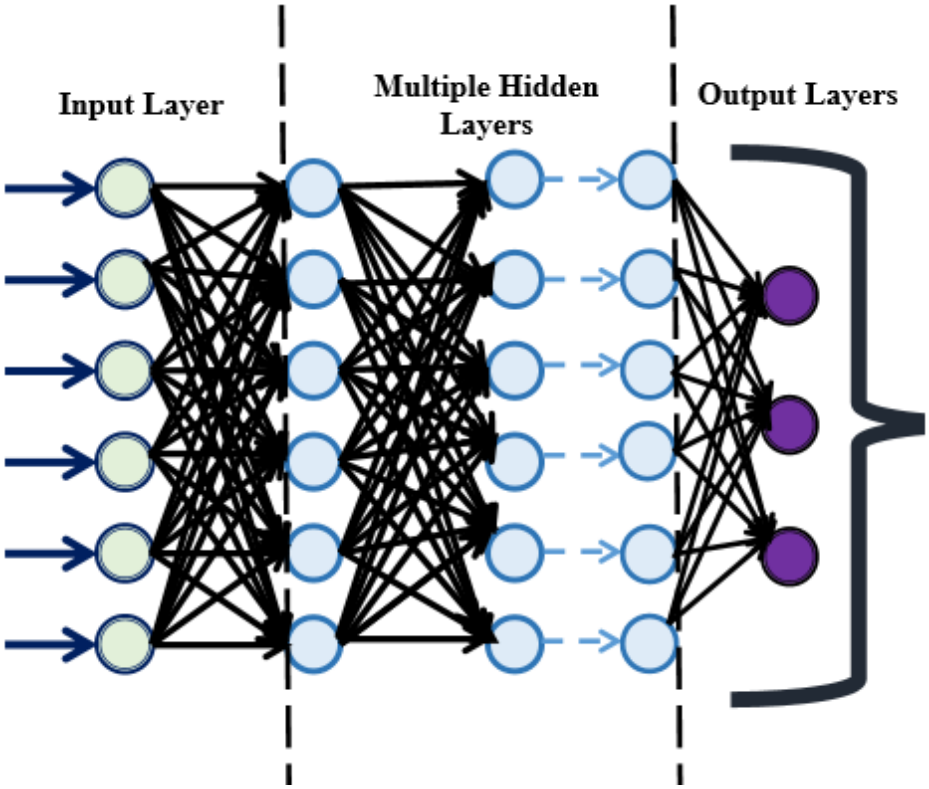


FIGURE 23. Multilayer Feedforward Neural Network (MLFF) used for gender prediction

describe its statistical properties. The MLP model maps the input features $\mathbf{x} \in \mathbb{R}^d$ to the output prediction vector $\hat{\mathbf{y}} \in \mathbb{R}^C$ using the transformation

$$\hat{\mathbf{y}} = f_o\left(W^{(2)} f_h\left(W^{(1)}\mathbf{x} + b^{(1)}\right) + b^{(2)}\right), \tag{17}$$

where $W^{(1)}$ and $W^{(2)}$ are the weight matrices, $b^{(1)}$ and $b^{(2)}$ are the bias vectors, $f_h(\cdot)$ denotes the hidden-layer activation function, and $f_o(\cdot)$ represents the output activation function. The trained classifier generates predictions $\hat{\mathbf{y}} = \arg\max_i \hat{y}_i$ for the given input images. The flowchart illustrating the alphabet recognition process is depicted in Fig. 24.

Different stages of alphabet recognition and Gender predictions from the handwritten prescriptions are shown below in Fig. 25.

4.4. **Symptoms prediction.** The EMNIST dataset, comprising 131,600 character images spanning 47 classes, is employed alongside a custom dataset derived from patient records. Handwritten gender image information is stored in `.csv` files for training and evaluation purposes. A Multilayer Perceptron (MLP) Neural Network serves as a classifier, with a Convolutional Neural Network (CNN) incorporated to enhance the classification process.

Let each input image be represented as $I(x, y) \in \mathbb{R}^{m \times n}$, where m and n indicate the image's height and width. The label associated with each image is represented

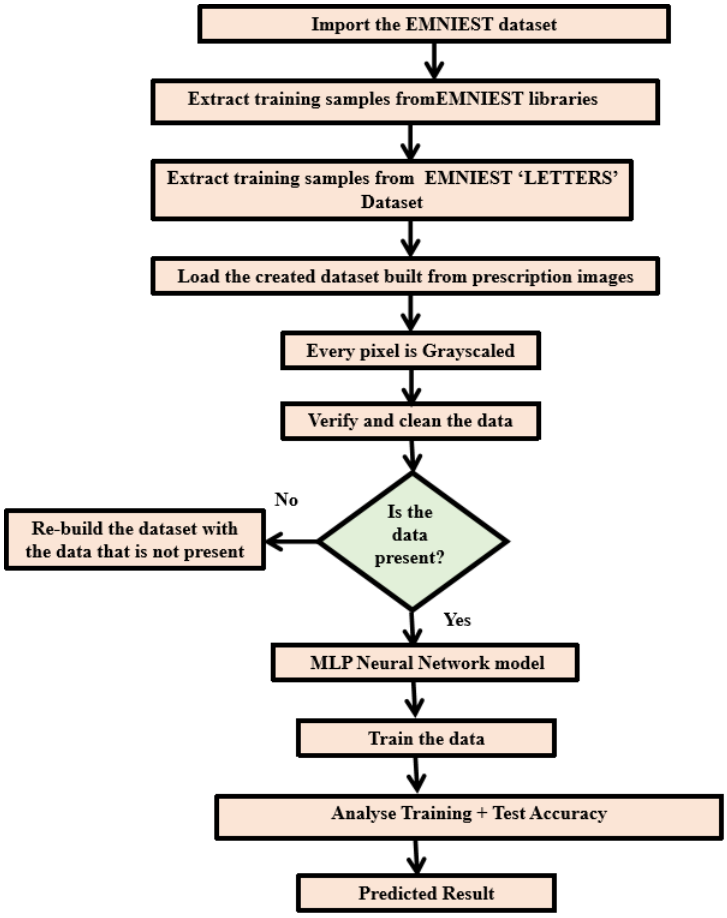


FIGURE 24. Flowchart illustrating the alphabet recognition process.

by $y \in \{1, 2, \ldots, C\}$, with $C = 47$ corresponding to the total number of classes. The MLP classifier maps the input feature vector $\mathbf{x} \in \mathbb{R}^d$ to the output prediction vector $\hat{\mathbf{y}} \in \mathbb{R}^C$ and is computed as

$$\hat{\mathbf{y}} = f_o\left(W^{(2)} f_h\left(W^{(1)}\mathbf{x} + b^{(1)}\right) + b^{(2)}\right), \tag{18}$$

where $W^{(1)}$ and $W^{(2)}$ represent the weight matrices for the first layer and second layer, respectively. The bias components $b^{(1)}$ and $b^{(2)}$ are added to the hidden and output layers. The function $f_h(\cdot)$ corresponds to the non-linear activation function applied within the hidden layer, and $f_o(\cdot)$ denotes the activation operation used to generate the final output. The trained classifier yields the predicted label as $\hat{y} = \arg\max_i \hat{y}_i$ for each input image. The flowchart describing the alphabet recognition procedure is presented in Fig. 24. In Fig. 26, the predicted symptoms of some diseases that are identified from the collected handwritten prescriptions are shown.

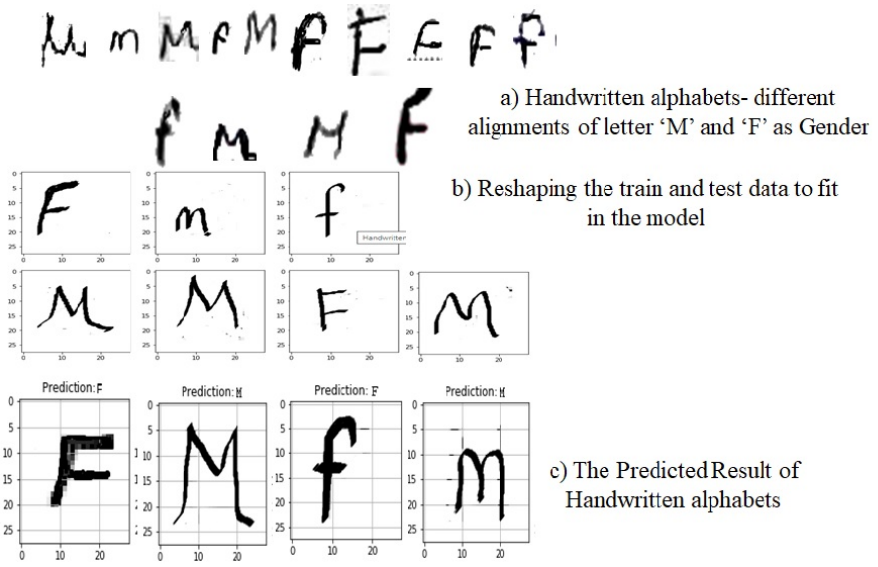


FIGURE 25. Different stages of alphabet recognition and Gender predictions from the handwritten prescriptions.

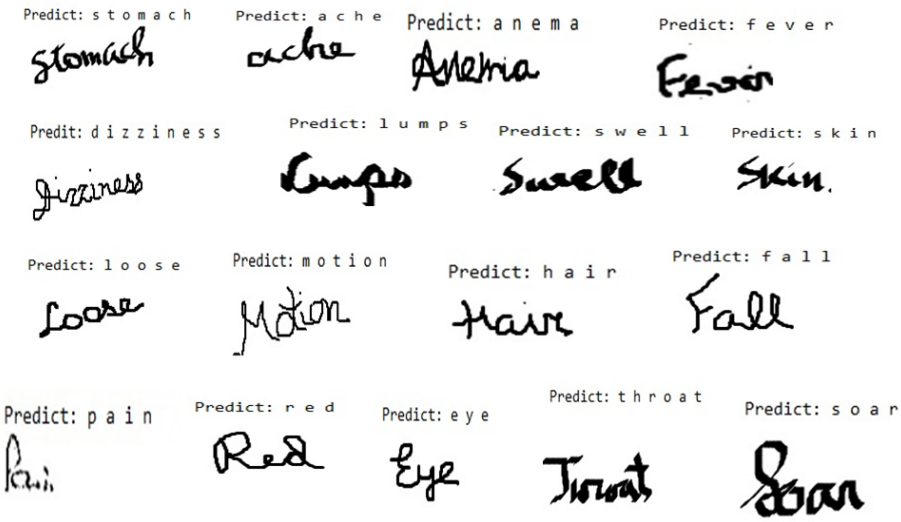


FIGURE 26. Predicted symptoms of some diseases identified from the collected handwritten prescriptions.

4.5. **Names of suggested medicines prediction.** Names of suggested Medicines Prediction: From the collected prescriptions, physicians typically recommend medications based on the observed symptoms corresponding to various diseases. When an immediate diagnosis is not feasible, additional diagnostic tests—such as blood, bone, saliva, urine, or stool analyses—are performed to identify the underlying condition. Following the test results, a definitive diagnosis is established, and the

appropriate medications are subsequently prescribed to the patient. Let the prescribed medicine set for patient i be represented as

$$M_i = \{m_1, m_2, \ldots, m_k\},$$

where each m_j denotes a specific medicine in the recommended treatment. Fig. 27 presents examples of predicted medicine names for selected cases.

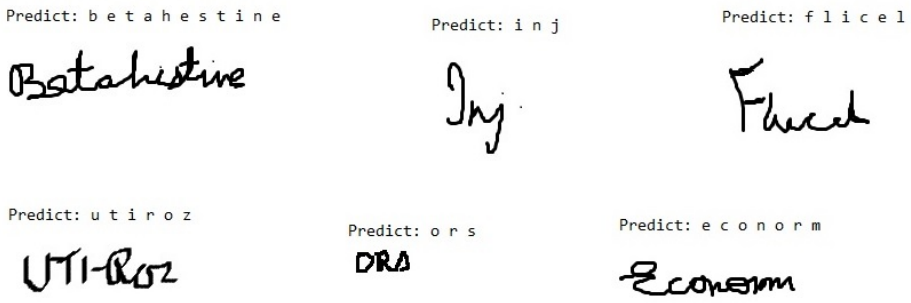


FIGURE 27. Names of suggested Medicines Prediction.

5. **Patient record generation.** The predicted outputs for each patient, including *Name*, *Age*, *Gender*, *Symptoms*, and *Medicines*, were aggregated to form a comprehensive digital patient record. This structured representation enables systematic storage, retrieval, and integration with electronic health record (EHR) systems.

For each patient, the digital record is defined as:

$$\mathbf{p}_i = \left[\text{Name}_i, \text{Gender}_i, \text{Age}_i, \text{Symptoms}_i, \mathcal{M}_i\right]$$

where

$$\mathcal{M}_i = \{m_1, m_2, \ldots, m_k\}$$

denotes the predicted set of prescribed medicines for the i-th patient.

5.1. **Deduction of each attribute from each prescription of each patient:** Our deep learning–based model was employed to predict handwritten alphabets and digits. In addition, the printed text present in each prescription was automatically recognized by the system.

Let the predicted handwritten sequence for patient i be represented as

$$H_i = [h_1, h_2, \ldots, h_n],$$

where h_j denotes the jth predicted character or digit. Similarly, the recognized printed text sequence is denoted as

$$P_i = [p_1, p_2, \ldots, p_m],$$

where p_k represents the kth printed character in the prescription.

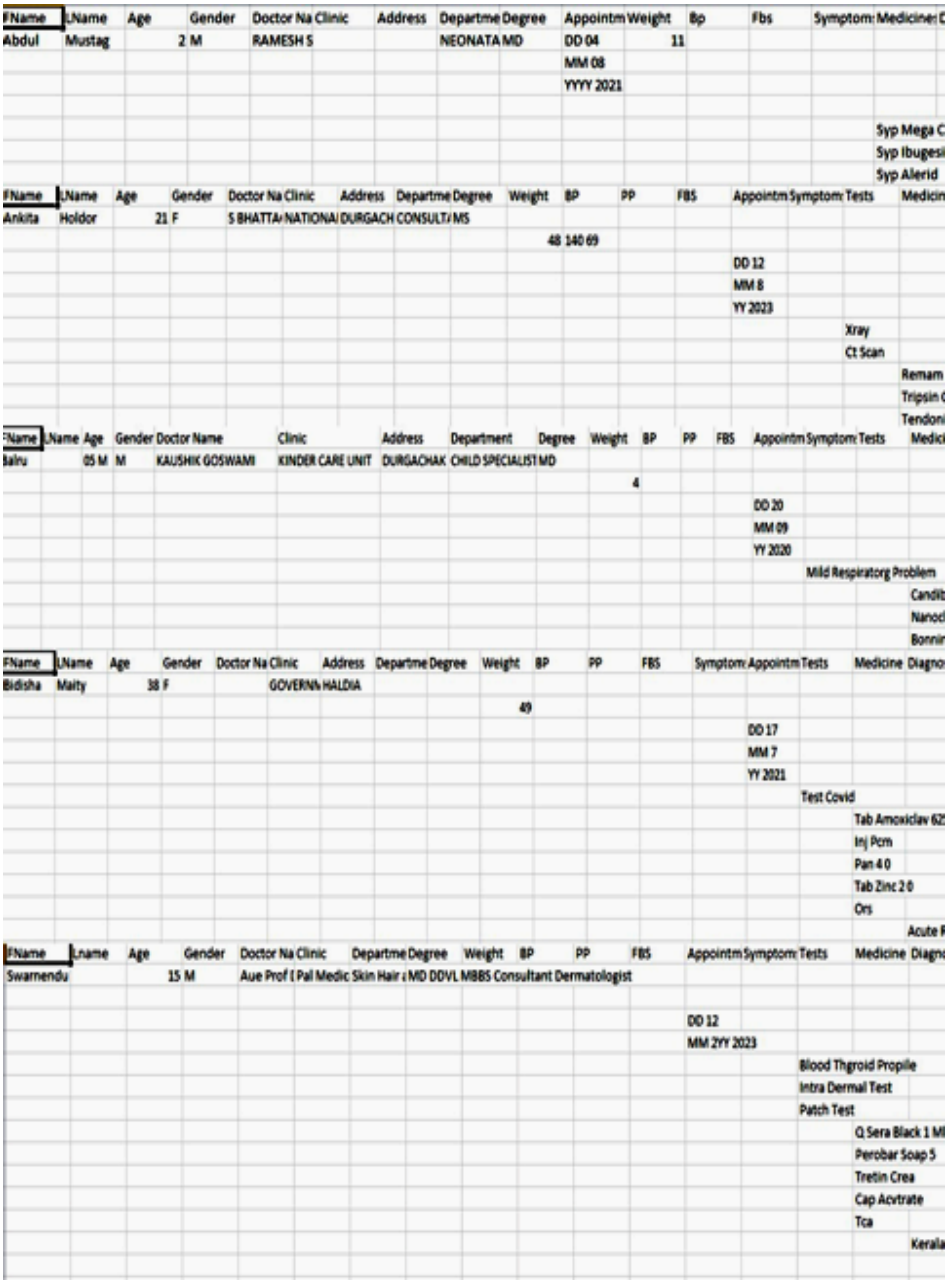
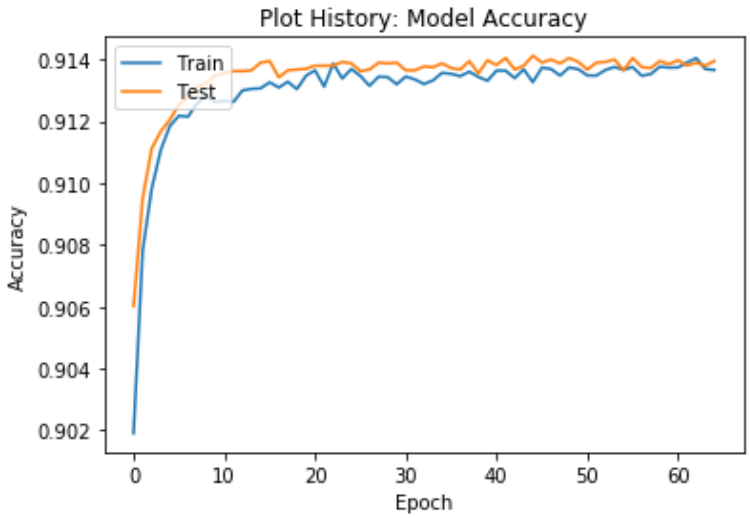


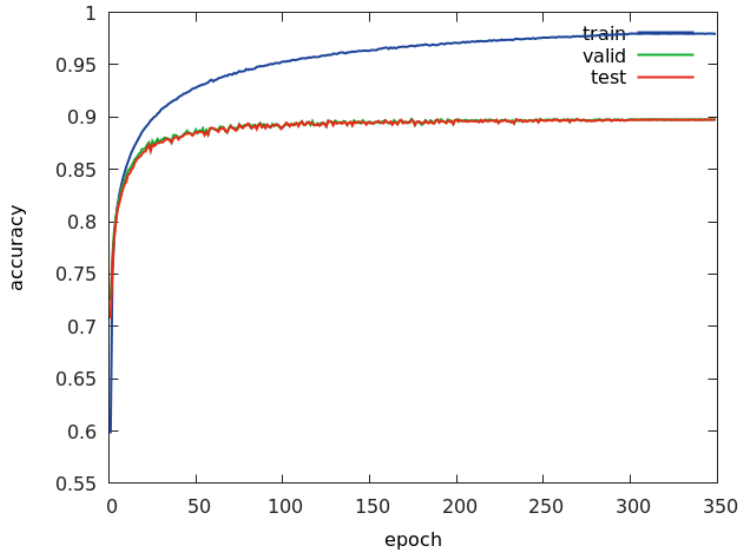
FIGURE 28. Handwritten to Digital record generation

6. **Results discussion.** Strong performance across a variety of prediction attributes confirms the efficacy of the proposed classifier for handwritten prescription recognition. As illustrated in Fig. 29, the system demonstrates high accuracy in predicting both *Name* and *Gender* attributes. The classifier achieves an accuracy of 91.2% for the Name attribute (Fig. 29a) and 88.6% for the Gender attribute (Fig. 29b). The superior performance in name recognition highlights the

model's ability to capture variations in letter formation and handwriting styles. The slightly lower gender classification accuracy may be attributed to inconsistent gender-denoting symbols or overlapping handwriting patterns.



(A) Name accuracy

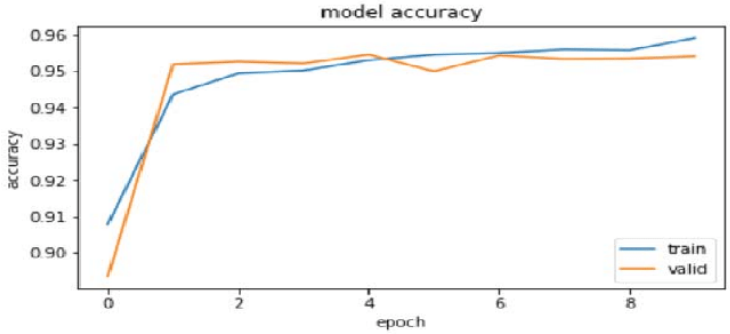


(B) Gender accuracy

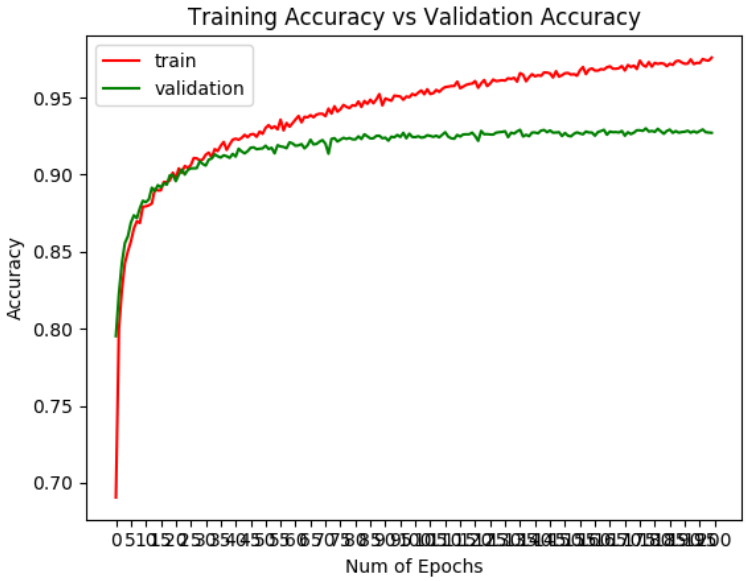
FIGURE 29. Name and gender attribute accuracy graphs

Similarly, Fig. 30 illustrates the effectiveness of the model in predicting age-related attributes. The model achieves an accuracy of 94.5% for single-digit age recognition (Fig. 30a) and 89.6% for multi-digit age recognition (Fig. 30b). The

reduced accuracy for multi-digit ages is primarily due to increased complexity arising from digit spacing irregularities, alignment distortions, and handwriting noise. Nevertheless, the model maintains consistent performance across numerically diverse datasets.



(A) Single-digit age accuracy

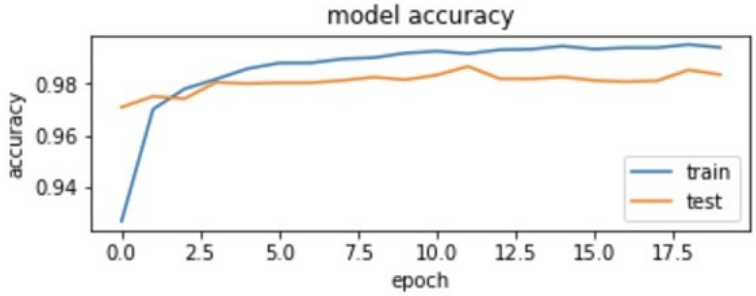


(B) Multi-digit age accuracy

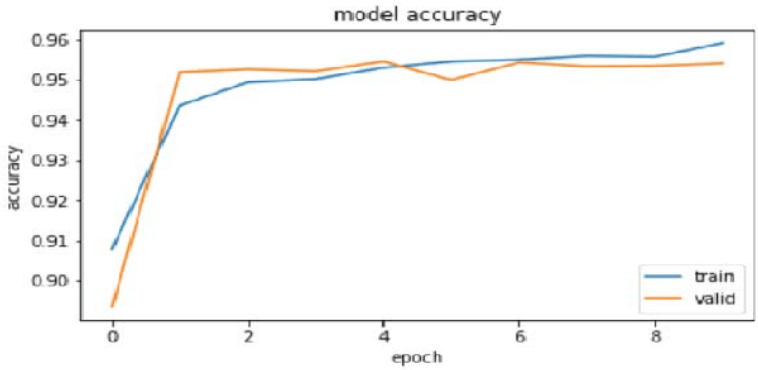
FIGURE 30. Age recognition accuracy graphs for single- and multi-digit ages

Figure 31 presents the results for medical attribute prediction. The classifier achieves an accuracy of 96.81% for symptom prediction and 95.01% for medicine prediction. Overall, the system correctly identifies 97.50% of symptoms and 95.50% of medicines. These results demonstrate the robustness of the proposed model in handling textual and contextual variations commonly found in handwritten prescriptions.

Overall, the results indicate that the CNN-based classifier consistently outperforms alternative approaches across all evaluated attributes. Despite challenges such



(A) Symptoms accuracy



(B) Medicine accuracy

FIGURE 31. Comparison of symptom and medicine prediction accuracy

as image noise, stroke overlap, and handwriting variability, the system maintains high predictive accuracy. These findings validate the proposed method as a reliable and practical solution for automated handwritten prescription interpretation in real-world healthcare digitization systems.

7. **Gap coverage analysis and benchmark comparison with current methods.** The proposed CNN–MLP-based framework was benchmarked against existing approaches for handwritten medical prescription recognition. Table 3 summarizes the major distinctions. The proposed method demonstrates notable improvements in recognition accuracy, computational efficiency, and scalability when compared with prior studies. Specifically, it:

- achieves a higher recognition accuracy of 90.20% on a large dataset containing over 400,000 handwritten prescriptions;
- utilizes all preprocessed samples to ensure comprehensive and unbiased model training;
- employs an optimized CNN–MLP hybrid architecture that maintains high accuracy while reducing computational complexity;
- operates independently of printed text components, enabling authentic handwritten text recognition;

TABLE 3. Comparative analysis of related works

Reference	**Limitations**	**Proposed Framework Improvements**
S. Tabassum *et al.*	Low recognition accuracy	Achieved 90.20% accuracy on a large-scale dataset
Esraa Hassan *et al.*	Small dataset and low test accuracy	Utilized over 400,000 prescriptions with significantly improved accuracy
P. Roy *et al.*	Dependent on printed text for keyword spotting	Eliminated dependency on printed text components
Seerat Rani *et al.*	Partial utilization of available data	Leveraged all preprocessed samples for complete training
E. Kamalanaban *et al.*	Complex and computationally expensive architecture	Reduced model complexity using an optimized CNN–MLP design
Raisi Z. *et al.*	Dataset inconsistencies	Collected and structured reliable clinical prescription data
Bajaj *et al.*	Overfitting in multilayer connector-based models	Mitigated overfitting via CNN–MLP hybridization
Sandhya Arora *et al.*	Evaluation limited to Devanagari script	Extended support to multiple languages using scalable MLP models

- supports multilingual recognition through multilayer perceptron (MLP) models trained on EMNIST and other language datasets, ensuring scalability, flexibility, and faster inference.

The modular hybrid architecture, combined with EMNIST-based transfer learning, enables robust feature extraction and rapid adaptation to previously unseen prescription data. Owing to its strong generalization capability, high recognition performance, and computational efficiency, the proposed framework is well suited for integration into Electronic Health Record (EHR) systems and real-world clinical workflows.

8. **Future work and conclusion.** This work presents a Deep Neural Network (DNN)-based Optical Character Recognition (OCR) framework for interpreting handwritten medical prescriptions. The system integrates Convolutional Neural Networks (CNNs) for spatial feature extraction, segmentation-guided text analysis, and an LSTM–CTC module for sequence modeling. This hybrid approach enables accurate patient-specific information extraction, improved readability, reliable digitization, and efficient healthcare data management.

The proposed framework achieves high recognition accuracy, with 91.2% accuracy for patient names, 96.8% for symptoms, 95.5% for medications, and 95.3% for age detection, demonstrating strong generalization across diverse handwriting styles. Furthermore, the structured `.csv` outputs ensure seamless integration with **Electronic Health Record (EHR)** systems, facilitating rapid data retrieval, clinical

analysis, and long-term record storage. This significantly reduces manual transcription errors and supports **AI-driven healthcare decision-making**.

Despite these advantages, certain limitations remain. The current system is restricted to English-language prescriptions and is dependent on the quality and diversity of the training dataset. Performance may degrade in the presence of highly inconsistent or overlapping handwriting. Additionally, the computational complexity of the framework limits its deployment on low-resource hardware, and handling sensitive medical data necessitates strict compliance with privacy and security regulations.

Several enhancements are planned as **future work**. Extending the dataset to include multilingual and region-specific prescriptions will improve generalization across diverse handwriting styles. Fuzzy decision-support methodologies, such as weighted Pythagorean fuzzy soft sets (WPFSS) [23], Sugeno–Weber triangular norms, and Complex Interval-Valued Intuitionistic Fuzzy Systems (CIVIFS) [23], can be integrated to enhance interpretability and decision reliability in uncertain clinical environments. Additional functionalities, including qualification keyword spotting, mobile and cloud deployment, doctor signature verification, and sentiment analysis for evaluating medication effectiveness, will further support real-time clinical applications.

The framework may also benefit from recent advances such as the rCTSFS with Dombi–Choquet MCDM algorithm proposed by Kakati *et al.* [24], which demonstrated superior diagnostic accuracy for diabetic retinopathy compared with numerous state-of-the-art approaches. Incorporating such techniques into an OCR-driven clinical framework can broaden applicability and further enhance recognition performance in complex medical scenarios.

In conclusion, the proposed framework represents a significant step toward intelligent, scalable, and secure healthcare automation. By combining handwritten prescription recognition with AI-based fuzzy decision-support systems such as WPFSS and rCTSFS with Dombi–Choquet integration, the framework enables safe, interpretable, and data-driven clinical decision-making under uncertainty. Future research will focus on integrating these advanced methodologies into existing experimental pipelines to further enhance clinical utility and diagnostic reliability.

Acknowledgments. The author used AI-based tools solely for grammar and formatting improvements; all scientific content and interpretations are original and human-authored. The author gratefully acknowledges the guidance and support of their co-authors and contributors, Dr. Bikash Kanti Sarkar, Dr. Abhijit Mustafi, and Dr. Apratim Mitra, throughout the preparation of this research work.

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Received June 9, 2025; revised November 12, 2025; early access February 6, 2026.